

WORK EXPERIENCE

Senior Software Engineer

Aug 2025 — Present

Intuit Inc

San Diego, CA

- Led the end-to-end implementation and launch of Pooled Routing for Quickbooks IVA group in CUP, which uses the Routing Decision Engine (RDE) to route each customer to the best-fit queue. Simplifies rule creation for operators by moving the heavy lifting of queue selection to RDE.
- Engineered the AI-driven IXP Playground platform, accelerating experimental setups from 3 months to 1 day and enabling the validation of a \$3.5M strategic routing initiative.
- Restored QuickBooks Midmarket ABR match rates from 0% to >80% by diagnosing systemic degradations via Splunk telemetry and Spout flow analysis; established automated anomaly detection for long-term stability.
- Led cross-functional diagnosis of critical Skill Sync bugs, doubling global match rates from 11% to 21% and unlocking an estimated \$5M in revenue through optimized expert matching logic.
- Developed a self-healing batch processor that reduced job runtime by 80% (60 mins to 12 mins) and shifted sync frequency from daily to hourly for 20k+ records per run.
- Orchestrated high-concurrency Tax Peak support, reducing customer wait times from 2 minutes to 5 seconds and automating the mapping of 12k experts across AWS Connect to drive 2,500+ daily optimal matches.

Software Engineer 2

Jan 2024 — Jul 2025

Intuit Inc

San Diego, CA

- Architected and deployed "Route Genie," a high-scale synchronization service integrating Expert Profile Service (EPS) with Amazon Connect; engineered critical Expert Credential and Capacity Sync features to optimize global routing.
- Developed a centralized CUP Admin UI dashboard for real-time skill white-listing, enhancing operational visibility; led rigorous "bug bash" initiatives resulting in a zero-defect production release.
- Engineered the "Demonstrated Skill Service" from the ground up using a Java-based "paved road" architecture; single-handedly designed and launched the service in 2 weeks to onboard 17,000 Product Experts for peak season.
- Spearheaded multi-region resiliency for the Contact Scheduler service, conducting deep-dive research into NoSQL, distributed caching, and EventBus architectures to support 10,000+ tax experts in TurboTax Live.
- Implemented high-throughput event processing by consuming appointment data via EventBus, ensuring 1:1 decoupled and Pooled ABR video appointments remained synchronized across global regions.
- Optimized Appointment Runtime monitoring by developing advanced Splunk dashboards and proactive alerting systems, ensuring 99.9% uptime during the critical FY24 launch.

Software Engineer

June 2023 — Dec 2023

Viasat Inc

Carlsbad, CA

- Mentored the interns in building the machine learning component of the modem self-healing system.
- Automated VS1 High Speed external router test scenarios which check the behavior of the modem when multiple external routers like Hitron, Netgear etc., are connected to it.
- Developed Health Monitoring framework for tracking the health of modems in AB and SB2 waveform.

Software Engineer

Aug 2020 — Jul 2021

Zomentum

Bangalore, India

- Developed and maintained data integration API for a third-party PSA tool called Connectwise Manage.
- Streamlined Git-Lab workflows which eased the deployment process.
- Contributed to the rule discovery phase for the integration of a payment gateway called Stripe with Zomentum.
- Participated in weekly bug triage sessions and made the application robust.

Software Engineer

Jun 2018 — Apr 2020

Soroco India Pvt Ltd

Bangalore, India

- Created Offer Letter Bot to automate the process of generating offer letters and save time for the HR team.
- Built several scalable and fault-tolerant end-to-end process automation systems in Python for an insurance firm.
- Wrote a configurable python script for Synthetic Data Generation which generated data for testing new features in Scout.
- Built an automation system for the Fin-Ops team of a client which handles sensitive user permission data from multiple sources like RDS, Redshift, S3 buckets, etc., where, I gained knowledge on using design patterns like Singleton and Factory methods.
- Worked with the Input engine team which uses Machine Learning to parse PDF files.
- Developed back-end API's in Django for a client-specific dashboard that fetched data from Amazon S3, RDS, and several other client-specific endpoints.

RESEARCH EXPERIENCE

Optimal Behavior Models for Shared Controls

Research Assistant - Advised by Prof. Sriram Sankaranarayanan

Spring 2023

Boulder, CO

- Developed a lane-following PID controller for Bicycle robot model.
- Conducted experiments with various scenarios for the following vehicle like controller-only, human-only and shared-driving modalities and observed the behavior pattern with respect to velocity and stopping distance.
- Z3 tool exploration to model obstacle avoiding use case.

Modelling Controllers for Cyber Physical Systems using Neural Networks

Independent Study - Advised by Prof. Sriram Sankaranarayanan

Fall 2022

Boulder, CO

- Investigated methods for successfully training neural networks in a control-based environment, such as trajectory tracking.
- Implemented Behavioral Cloning and Data Set Aggregation (DaGGER), two imitation learning techniques, to efficiently train neural networks.
- Conducted tests and evaluated the effectiveness of the aforementioned strategies using the Bicycle robot and Caltech Ducted Fan benchmarks.

Brain Tumor Segmentation (BraTS)

Independent Study - Advised by Prof. Geena Kim

Spring 2022

Boulder, CO

- As part of the BraTS challenge, explored novel techniques and techniques for effectively segmenting brain tumors.
- In-depth examination of the reasons why the current Deep InfoMax strategy failed to outperform a traditional U-net-based architecture.
- Adopted the multi-scale dense U-net (MDU-net) architecture, which produced encouraging outcomes.

TEACHING EXPERIENCE

Teaching Assistant

CSCI 3155 - Principles of Programming Languages Instructor: Prof. Sriram Sankaranarayanan

Spring 2023

Teaching Assistant

CSCI 3155 - Principles of Programming Languages Instructor: Prof. Bor Yuh Evan Chang

Fall 2022

Teaching Assistant

CSCI 3155 - Principles of Programming Languages Instructor: Prof. Sriram Sankaranarayanan

Spring 2022

PROJECTS

Task Offloading using Edge Computing

CSCI 5673 - Distributed Systems

Spring 2023

Boulder, CO

- Developed a video processing application which uses Collaborative Offloading technique in Edge computing to efficiently process input videos from various edge devices like Raspberry Pi, Android mobile etc.,
- Performed an evaluative study using several heuristics and observed the behavior and efficiency of video processing.

Restaurant Delivery

CSCI 5854 - Theoretical Foundations of Autonomous Systems

Fall 2022

Boulder, CO

- Used the Omega Threads tool to create controllers from an LTL specification. A bicycle's dynamics and basic drone dynamics were used to produce two distinct delivery modalities.

Music Generation

CSCI 5922 - Neural Networks & Deep Learning

Spring 2022

Boulder, CO

- Performed a comparative study on the quality of music generated by different deep learning models like RNN, LSTM, GRU, and Distill-GPT2 with over **1000** musical notes(ABC notation) obtained from Nottingham music database.

Sudoku Solver

CSCI 5622 - Machine Learning

Fall 2021

Boulder, CO

- Developed a Convolutional Neural Network (CNN) model that solves a 9x9 Sudoku puzzle one cell at a time with an accuracy of 100%

EDUCATION

Master of Science in Computer Science, University of Colorado Boulder CGPA 3.94

Aug 2021 - May 2023

Bachelor of Engineering in Computer Science, Coimbatore Institute of Technology. CGPA 9.04

Jul 2014 - May 2018

COURSE WORK

Machine Learning, NLP, Data Mining, Linear Programming, Neural Networks and Deep Learning, Theoretical Foundations of Autonomous Systems, Distributed Systems

SKILLS

Programming Languages	Python, Scala, C/C++, Java, React
Databases	PostgreSQL, MongoDB, Druid
Frameworks	Django, Play, gRPC, SOAP, Spring Boot
Technology	Git, AWS, Jenkins, Grafana, Docker, Splunk, Redis

CERTIFICATIONS

- Scala and Functional programming for beginners.
- Advanced Scala and Functional Programming.
- Docker Essentials training

ACHIEVEMENTS

Received Outstanding Teaching Assistant award for my contributions in CSCI 3155
Secretary of National Service Scheme Club in CIT
Event Manager of a technical symposium named Interface'16
Bestowed with Best Student Teacher Award